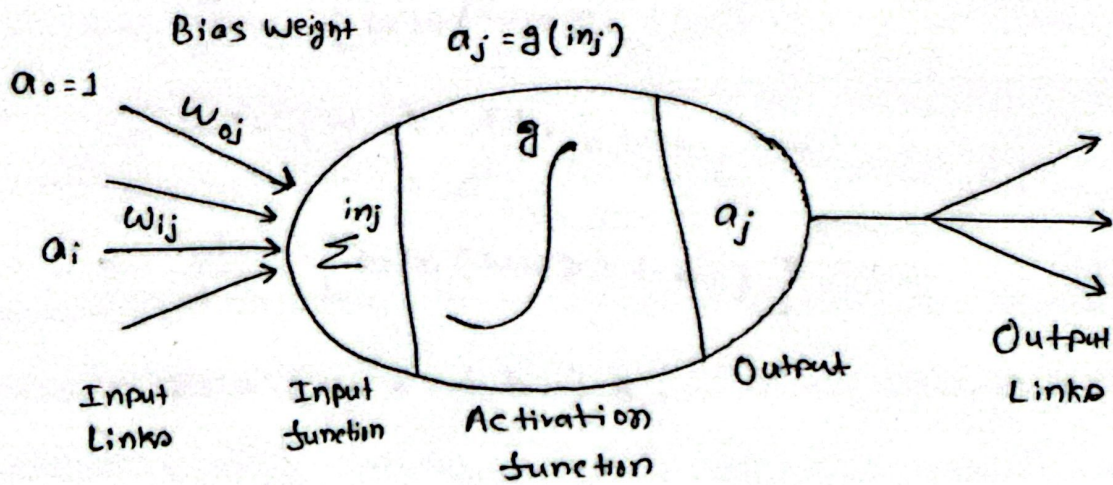


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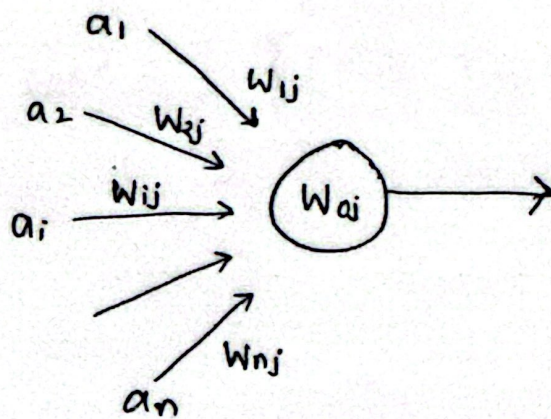
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$$a_j = g \left(\sum_{i=0}^n w_{i,j} a_i \right)$$

a_i = output activation of unit i

$w_{i,j}$ = weight on the link



$$In = a_1 w_{1j} + a_2 w_{2j} + \dots + a_i w_{ij} + \dots + a_n w_{nj}$$

$$a_j = g(In_j)$$

$$In_j = \sum_{i=0}^n w_{i,j} a_i$$

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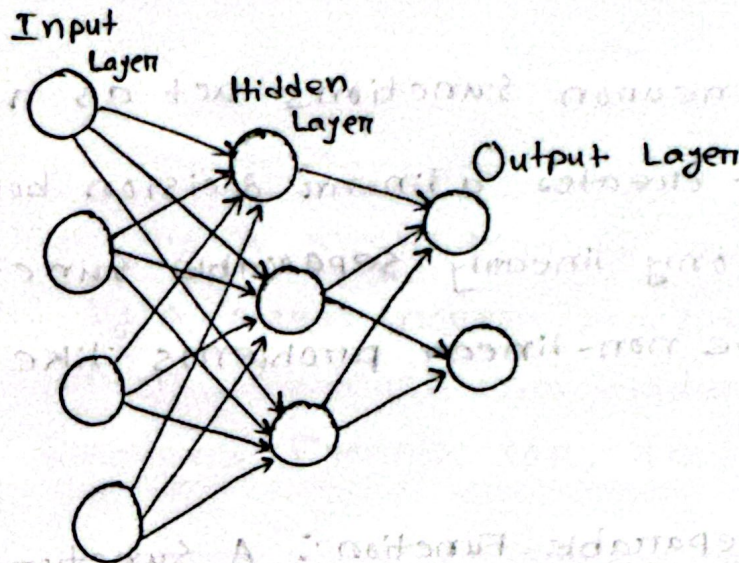
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2 type of Neural Network.

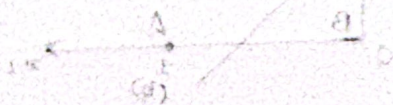
i) Feed Forward Neural Network

ii) Recurrent Neural Network



Feedforward Neural Network:

- i) A type of artificial network where information moves in one direction only. (input $\rightarrow$ hidden layer(s) $\rightarrow$ output)
- ii) No cycles or feedback loops between neurons.
- iii) consists of three main parts (input, hidden, output)
- iv) each neuron: receives weighted inputs, applies a weighted sum, passes the result through an activation function.
- v) common activation function: ReLU, sigmoid, Tanh, softmax



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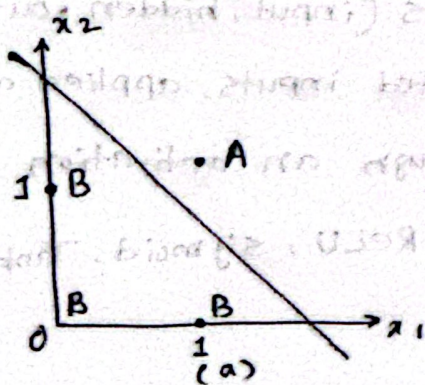
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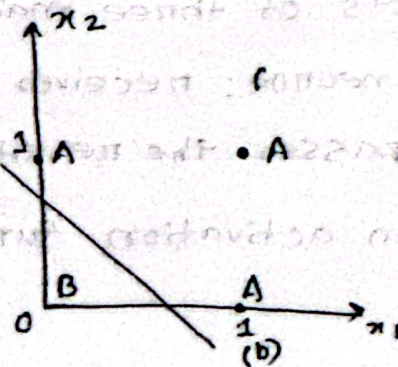
Recurrent Neural Network: A type of neural network designed for sequential data. Its output depends on input and previous state. It's suitable for short term memory. Feedback connection is available.

A single neuron functions act as a linear classifier because it creates a linear decision boundary, so it can represent only linearly separable functions and cannot solve non-linear problems like XOR.

Linearly Separable Function: A function is linearly separable if its input classes can be separated using a single straight line. All points of one class lie on one side of the boundary and the other class lies on the opposite side. It can be implemented by a single neuron. Example: AND, OR.



AND



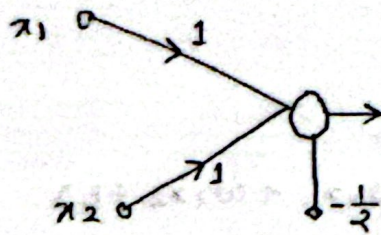
OR

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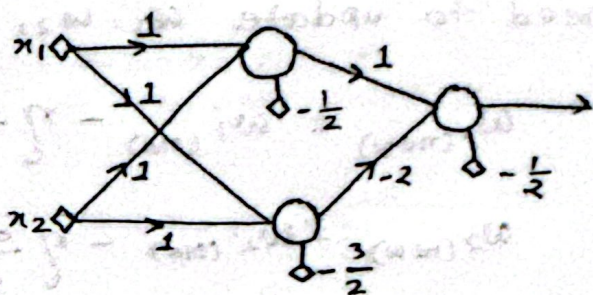
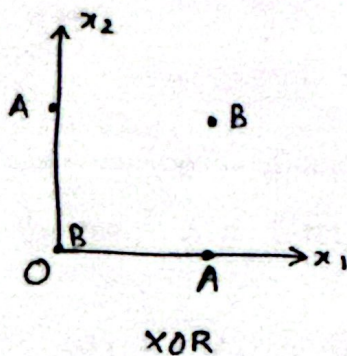
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The OR gate implementation:



If you change the bias to -1.5 , it becomes an AND gate.

Linearly Non Separable function: If its input classes cannot be separated by a single straight line. It cannot be implemented by a single neuron/perceptron. It requires: multiple neurons, at least one hidden layer, nonlinear activation functions. Example: XOR, XNOR.

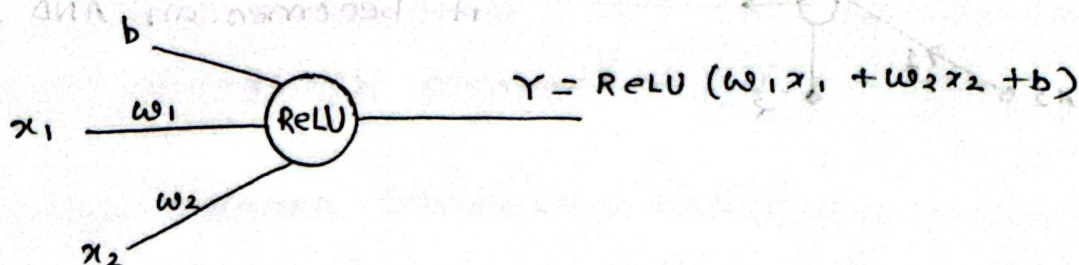


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Parameter Update with ReLU Function



The loss function,

$$\text{Loss} = \frac{1}{2} (O_b - Y)^2$$

O_b = actual target

Y = Predicted target

To train this neural network model, we have to continuously update the parameters weights and biases. Here, we need to update w_1, w_2, b .

$$w_{1(\text{new})} = w_{1(\text{old})} - \eta \frac{d\text{Loss}}{dw_1}$$

$$w_{2(\text{new})} = w_{2(\text{old})} - \eta \frac{d\text{Loss}}{dw_2}$$

$$b_{(\text{new})} = b_{(\text{old})} - \eta \frac{d\text{Loss}}{db}$$

So, we need to differentiate loss function $\text{Loss} = \frac{1}{2} (O_b - Y)^2$ in terms of w_1 and w_2, b .

$$Y = \text{Relu}(x_1 w_1 + x_2 w_2 + b)$$

$$\frac{d\text{Loss}}{dw_1} = \frac{d\text{Loss}}{dY} \times \frac{dY}{dw_1}$$

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$$\frac{dLoss}{dw_2} = \frac{dLoss}{dY} \times \frac{dY}{dw_2}$$

$$\frac{dLoss}{db} = \frac{dLoss}{dY} \times \frac{dY}{db}$$

Now,

$$\begin{aligned} \frac{dLoss}{dY} &= \frac{d}{dY} \left\{ \frac{1}{2} (0b - Y)^2 \right\} \\ &= \frac{1}{2} \times 2 (0b - Y) (-1) = -(0b - Y) = -Error \end{aligned}$$

$$\begin{aligned} \frac{dY}{dw_1} &= \frac{d}{dw_1} [\text{ReLU}(w_1 x_1 + w_2 x_2 + b)] \\ &= \text{ReLU}'(w_1 x_1 + w_2 x_2 + b) x_1 \end{aligned}$$

$$\begin{aligned} \frac{dLoss}{dw_1} &= \frac{dLoss}{dY} \times \frac{dY}{dw_1} \\ &= -Error \times \text{ReLU}'(w_1 x_1 + w_2 x_2 + b) x_1 \\ &= -\delta x_1 \end{aligned}$$

here, $\delta = Error \times \text{ReLU}'(w_1 x_1 + w_2 x_2 + b)$

$$\begin{aligned} \therefore w_1 &= w_1 - \eta \frac{dLoss}{dw_1} \\ &= w_1 - \eta (-\delta x_1) = \boxed{w_1 + \eta \delta x_1} \end{aligned}$$

$$\frac{dY}{dw_2} = \frac{d}{dw_2} [\text{ReLU}(w_1 x_1 + w_2 x_2 + b)]$$

$$= \text{ReLU}'(w_1 x_1 + w_2 x_2 + b) x_2$$

$$\begin{aligned} \frac{dLoss}{dw_2} &= -Error \times \text{ReLU}'(w_1 x_1 + w_2 x_2 + b) x_2 \\ &= -\delta x_2 \end{aligned}$$

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$$w_2 = w_2 - \eta \frac{d\text{Loss}}{dw_2}$$

$$= w_2 - \eta (-x_2 \delta) = \boxed{w_2 + \eta \delta x_2}$$

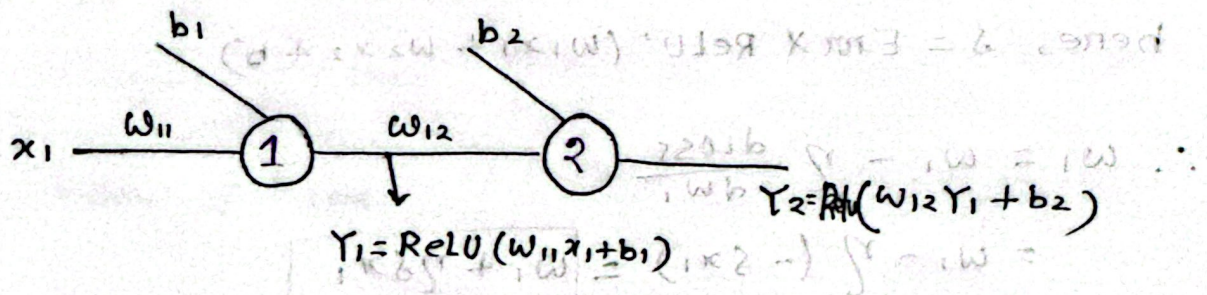
$$\frac{dY}{db} = \frac{d}{db} [\text{ReLU}(w_1 x_1 + w_2 x_2 + b)]$$

$$= \text{ReLU}'(w_1 x_1 + w_2 x_2 + b) \times 1$$

$$\frac{d\text{Loss}}{db} = -\text{Err} \times \text{ReLU}'(w_1 x_1 + w_2 x_2 + b) = -\delta$$

$$b = b - \eta \frac{d\text{Loss}}{db} = b - \eta (-\delta) = \boxed{b + \eta \delta}$$

Let us have another Neural Network:



Here, weights are w_{11} and w_{12} and biases are b_1, b_2 .

$$\text{Here Loss} = \frac{1}{2} (ob_2 - y_2)^2$$

We need to update w_{12} and b_2 first for neuron 2.

$$w_{12} = w_{12} - \eta \frac{d\text{Loss}}{dw_{12}}$$

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$$\frac{d\text{Loss}}{dw_{12}} = \frac{d\text{Loss}}{dy_2} \times \frac{dy_2}{dw_{12}}$$

$$\frac{d\text{Loss}}{dy_2} = \frac{d}{dy_2} \left\{ \frac{1}{2} (0b_2 - y_2)^2 \right\}$$

$$= \frac{1}{2} \times 2 (0b_2 - y_2) \times (-1) = -(0b_2 - y_2) = -\text{Err}_2$$

$$\frac{dy_2}{dw_{12}} = \frac{d}{dw_{12}} \text{ReLU}(w_{12}y_1 + b_2)$$

$$= \text{ReLU}'(w_{12}y_1 + b_2) \times y_1$$

$$\therefore w_{12} = w_{12} - \eta \frac{d\text{Loss}}{dw_{12}}$$

$$= w_{12} - \eta (-\text{Err}_2 \times \text{ReLU}'(w_{12}y_1 + b_2) y_1)$$

$$w_{12} = w_{12} + \eta \delta_2 y_1$$

Similarly for b_2 ,

$$\frac{d\text{Loss}}{db_2} = \frac{d\text{Loss}}{dy_2} \times \frac{dy_2}{db_2}$$

$$= -\text{Err}_2 \times \frac{d}{db_2} [\text{ReLU}(w_{12}y_1 + b_2)]$$

$$= -\text{Err}_2 \times \text{ReLU}'(w_{12}y_1 + b_2) = -\delta_2$$

$$\therefore b_2 = b_2 - \eta \frac{d\text{Loss}}{db_2} = b_2 - \eta (-\delta_2) = b_2 + \eta \delta_2$$

Now, we need to update w_{11} and b_1 for neuron 1.

$$\text{So, } w_{11} = w_{11} - \eta \frac{d\text{Loss}}{dw_{11}}$$

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$$\frac{dLoss}{dw_{11}} = \frac{dLoss}{dY_2} \cdot \frac{dY_2}{dw_{11}}$$

$$= \frac{dLoss}{dY_2} \cdot \frac{dY_2}{dY_1} \cdot \frac{dY_1}{dw_{11}}$$

$$\frac{dLoss}{dY_2} = -Err_2$$

$$\frac{dY_2}{dY_1} = \frac{d}{dY_1} [ReLU(W_{12}Y_1 + b_2)]$$

$$= ReLU'(W_{12}Y_1 + b_2)W_{12}$$

$$\frac{dY_1}{dw_{11}} = \frac{d}{dw_{11}} [ReLU(W_{11}x_1 + b_1)]$$

$$= ReLU'(W_{11}x_1 + b_1)x_1$$

$$\therefore W_{11} = W_{11} - \eta (-Err_2 \times ReLU'(W_{12}Y_1 + b_2)W_{12})$$

$$\times ReLU'(W_{11}x_1 + b_1)x_1$$

$$= W_{11} + \eta \delta_2 W_{12} \times ReLU'(W_{11}x_1 + b_1)x_1$$

Similarly we will get,

$$b_1 = b_1 - \eta \frac{dLoss}{db_1}$$

$$= b_1 - \eta \left(\frac{dLoss}{db_1} \times \frac{dY_2}{dY_1} \times \frac{dY_1}{db_1} \right)$$

$$= b_1 - \eta (-Err_2 \times ReLU'(W_{12}Y_1 + b_2)W_{12}) \times ReLU'(W_{11}x_1 + b_1)$$

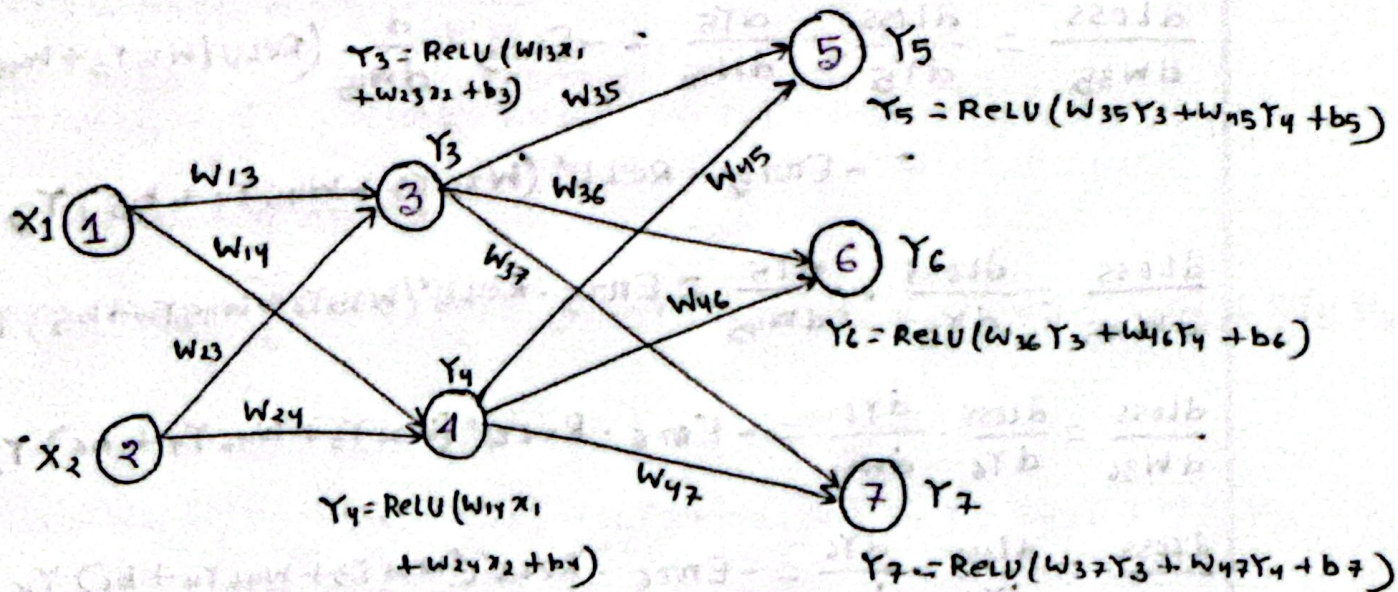
$$= b_1 + \eta \delta_2 W_{12} ReLU'(W_{11}x_1 + b_1)$$

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$$\text{Loss} = \frac{1}{2} \left\{ (Ob_5 - Y_5)^2 + (Ob_6 - Y_6)^2 + (Ob_7 - Y_7)^2 \right\}$$

We need to start with output nodes.

$$\frac{d\text{Loss}}{dY_5} = \frac{d}{dY_5} \left\{ \frac{1}{2} \times 2 (Ob_5 - Y_5) (-1) \right\} = -Err_5$$

$$\frac{d\text{Loss}}{dY_6} = \frac{d}{dY_6} \left\{ \frac{1}{2} \times 2 (Ob_6 - Y_6) (-1) \right\} = -Err_6$$

$$\frac{d\text{Loss}}{dY_7} = \frac{d}{dY_7} \left\{ \frac{1}{2} \times 2 (Ob_7 - Y_7) (-1) \right\} = -Err_7$$

$$Y_5 = \text{ReLU}(W_{35}Y_3 + W_{45}Y_4 + b_5)$$

$$Y_6 = \text{ReLU}(W_{36}Y_3 + W_{46}Y_4 + b_6)$$

$$Y_7 = \text{ReLU}(W_{37}Y_3 + W_{47}Y_4 + b_7)$$

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$$\frac{dLoss}{dW_{35}} = \frac{dLoss}{dY_5} \cdot \frac{dY_5}{dW_{35}} = -Err_5 \cdot \frac{d}{dW_{35}} (ReLU(W_{35}Y_3 + W_{45}Y_4 + b_5))$$

$$= -Err_5 \cdot ReLU'(W_{35}Y_3 + W_{45}Y_4 + b_5) Y_3$$

$$\frac{dLoss}{dW_{45}} = \frac{dLoss}{dY_5} \cdot \frac{dY_5}{dW_{45}} = Err_5 \cdot ReLU'(W_{35}Y_3 + W_{45}Y_4 + b_5) Y_4$$

$$\frac{dLoss}{dW_{36}} = \frac{dLoss}{dY_6} \cdot \frac{dY_6}{dW_{36}} = -Err_6 \cdot ReLU'(W_{36}Y_3 + W_{46}Y_4 + b_6) Y_3$$

$$\frac{dLoss}{dW_{46}} = \frac{dLoss}{dY_6} \cdot \frac{dY_6}{dW_{46}} = -Err_6 \cdot ReLU'(W_{36}Y_3 + W_{46}Y_4 + b_6) Y_4$$

$$\frac{dLoss}{dW_{37}} = \frac{dLoss}{dY_7} \cdot \frac{dY_7}{dW_{37}} = -Err_7 \cdot ReLU'(W_{37}Y_3 + W_{47}Y_4 + b_7) Y_3$$

$$\frac{dLoss}{dW_{47}} = \frac{dLoss}{dY_7} \cdot \frac{dY_7}{dW_{47}} = -Err_7 \cdot ReLU'(W_{37}Y_3 + W_{47}Y_4 + b_7) Y_4$$

$$\frac{dLoss}{db_5} = \frac{dLoss}{dY_5} \cdot \frac{dY_5}{db_5} = -Err_5 \cdot ReLU'(W_{35}Y_3 + W_{45}Y_4 + b_5)$$

$$\frac{dLoss}{db_6} = \frac{dLoss}{dY_6} \cdot \frac{dY_6}{db_6} = -Err_6 \cdot ReLU'(W_{36}Y_3 + W_{46}Y_4 + b_6)$$

$$\frac{dLoss}{db_7} = \frac{dLoss}{dY_7} \cdot \frac{dY_7}{db_7} = -Err_7 \cdot ReLU'(W_{37}Y_3 + W_{47}Y_4 + b_7)$$

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$$W_{35} = W_{35} - \eta \frac{dLoss}{dW_{35}}$$

$$= W_{35} - \eta (-Err_5 \cdot \text{ReLU}'(W_{35}Y_3 + W_{45}Y_4 + b_5)Y_3)$$

$$= W_{35} + \eta \delta_5 Y_3$$

$$W_{45} = W_{45} + \eta Err_5 \cdot \text{ReLU}'(W_{35}Y_3 + W_{45}Y_4 + b_5)Y_4 = W_{45} + \eta \delta_5 Y_4$$

$$W_{36} = W_{36} + \eta Y_3 \cdot Err_6 \cdot \text{ReLU}'(W_{36}Y_3 + W_{46}Y_4 + b_6) = W_{36} + \eta \delta_6 Y_3$$

$$W_{46} = W_{46} + \eta Err_6 \cdot \text{ReLU}'(W_{36}Y_3 + W_{46}Y_4 + b_6)Y_4 = W_{46} + \eta \delta_6 Y_4$$

$$W_{37} = W_{37} + \eta Y_3 \cdot Err_7 \cdot \text{ReLU}'(W_{37}Y_3 + W_{47}Y_4 + b_7) = W_{37} + \eta \delta_7 Y_3$$

$$W_{47} = W_{47} + \eta Err_7 \cdot \text{ReLU}'(W_{37}Y_3 + W_{47}Y_4 + b_7)Y_4 = W_{47} + \eta \delta_7 Y_4$$

$$b_5 = b_5 + \eta Err_5 \cdot \text{ReLU}'(W_{35}Y_3 + W_{45}Y_4 + b_5) = \eta \delta_5 + b_5$$

$$b_6 = b_6 + \eta Err_6 \cdot \text{ReLU}'(W_{36}Y_3 + W_{46}Y_4 + b_6) = \eta \delta_6 + b_6$$

$$b_7 = b_7 + \eta Err_7 \cdot \text{ReLU}'(W_{37}Y_3 + W_{47}Y_4 + b_7) = \eta \delta_7 + b_7$$

Now, we need to consider the hidden nodes 3 and 4.

$$Y_3 = \text{ReLU} \{ W_{13} \cdot X_1 + W_{23} X_2 + b_3 \}$$

$$Y_4 = \text{ReLU} \{ W_{14} \cdot X_1 + W_{24} X_2 + b_4 \}$$

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Now we need to differentiate the Loss function with respect to hidden outputs Y_3 and Y_4 . Both nodes 3 and 4 have parents 5, 6, 7.

$$\frac{dLoss}{dY_3} = \frac{dLoss}{dY_5} \cdot \frac{dY_5}{dY_3} + \frac{dLoss}{dY_6} \cdot \frac{dY_6}{dY_3} + \frac{dLoss}{dY_7} \cdot \frac{dY_7}{dY_3}$$

$$= (-\delta_5) W_{35} + (-\delta_6) W_{36} + (-\delta_7) W_{37}$$

$$\frac{dLoss}{dY_4} = \frac{dLoss}{dY_5} \cdot \frac{dY_5}{dY_4} + \frac{dLoss}{dY_6} \cdot \frac{dY_6}{dY_4} + \frac{dLoss}{dY_7} \cdot \frac{dY_7}{dY_4}$$

$$= (-\delta_5) W_{45} + (-\delta_6) W_{46} + (-\delta_7) W_{47}$$

$$\frac{dLoss}{dW_{13}} = \frac{dLoss}{dY_3} \cdot \frac{dY_3}{dW_{13}} = \{(-\delta_5) W_{35} + (-\delta_6) W_{36} + (-\delta_7) W_{37}\}$$

$$\text{ReLU}'(W_{13} \cdot x_1 + W_{23} \cdot x_2 + b_3) x_1$$

$$\frac{dLoss}{dW_{23}} = \frac{dLoss}{dY_3} \cdot \frac{dY_3}{dW_{23}} = \{(-\delta_5) W_{35} + (-\delta_6) W_{36} + (-\delta_7) W_{37}\}$$

$$\text{ReLU}'(W_{13} \cdot x_1 + W_{23} \cdot x_2 + b_3) x_2$$

$$\frac{dLoss}{dW_{14}} = \frac{dLoss}{dY_4} \cdot \frac{dY_4}{dW_{14}} = \{(-\delta_5) W_{45} + (-\delta_6) W_{46} + (-\delta_7) W_{47}\}$$

$$\text{ReLU}'(W_{14} \cdot x_1 + W_{24} \cdot x_2 + b_4) x_1$$

$$\frac{dLoss}{dW_{24}} = \frac{dLoss}{dY_4} \cdot \frac{dY_4}{dW_{24}} = \{(-\delta_5) W_{45} + (-\delta_6) W_{46} + (-\delta_7) W_{47}\}$$

$$\text{ReLU}'(W_{14} x_1 + W_{24} x_2 + b_4) x_2$$

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$$\frac{dLoss}{db_3} = \frac{dLoss}{dY_3} \frac{dY_3}{db_3} = \{(-\delta_5)W_{35} + (-\delta_6)W_{36} + (-\delta_7)W_{37}\} \text{ReLU}'(W_{13}X_1 + W_{23}X_2 + b_3)$$

$$\frac{dLoss}{db_4} = \frac{dLoss}{dY_4} \frac{dY_4}{db_4} = \{(-\delta_5)W_{45} + (-\delta_6)W_{46} + (-\delta_7)W_{47}\} \text{ReLU}'(W_{14}X_1 + W_{24}X_2 + b_4)$$

$$W_{13} = W_{13} + \eta(\delta_5 W_{35} + \delta_6 W_{36} + \delta_7 W_{37}) \text{ReLU}'(W_{13}X_1 + W_{23}X_2 + b_3)X_1$$

$$W_{23} = W_{23} + \eta(\delta_5 W_{35} + \delta_6 W_{36} + \delta_7 W_{37}) \text{ReLU}'(W_{13}X_1 + W_{23}X_2 + b_3)X_2$$

$$W_{14} = W_{14} + \eta(\delta_5 W_{45} + \delta_6 W_{46} + \delta_7 W_{47}) \text{ReLU}'(W_{14}X_1 + W_{24}X_2 + b_4)X_1$$

$$W_{24} = W_{24} + \eta(\delta_5 W_{45} + \delta_6 W_{46} + \delta_7 W_{47}) \text{ReLU}'(W_{14}X_1 + W_{24}X_2 + b_4)X_2$$

$$b_3 = b_3 + \eta(\delta_5 W_{35} + \delta_6 W_{36} + \delta_7 W_{37}) \text{ReLU}'(W_{13}X_1 + W_{23}X_2 + b_3)$$

$$b_4 = b_4 + \eta(\delta_5 W_{45} + \delta_6 W_{46} + \delta_7 W_{47}) \text{ReLU}'(W_{14}X_1 + W_{24}X_2 + b_4)$$

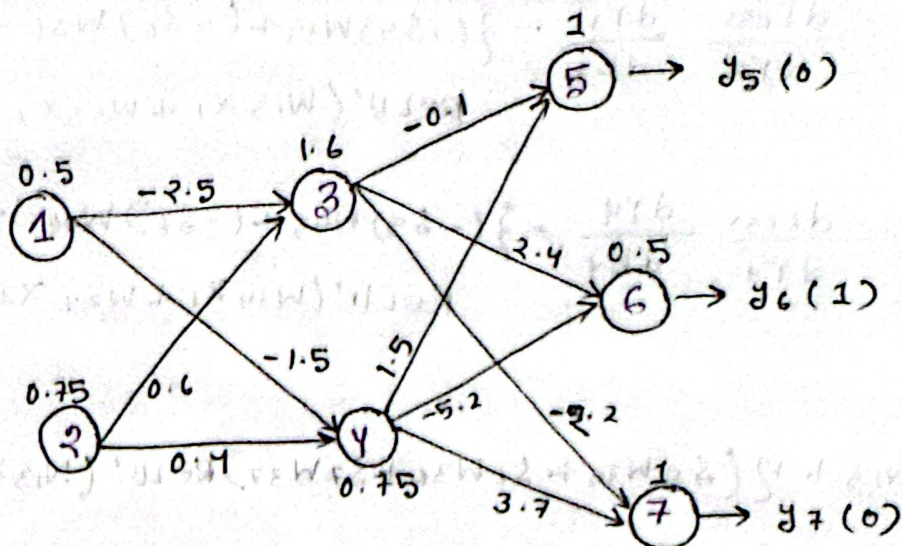
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$$y_3' = \text{ReLU}(x_1 w_{13} + x_2 w_{23} + b_3) = \text{ReLU}(-0.5 \times 2.5 + 0.75 \times 0.6 + 1.6) = 0.8$$

$$y_4' = \text{ReLU}(-0.5 \times 1.5 + 0.75 \times 0.4 + 0.7) = 0.3$$

$$y_5' = \text{ReLU}(-0.1 \times 0.8 + 0.3 \times 1.5 + 1) = 1.37$$

$$y_6' = \text{ReLU}(0.8 \times 2.4 - 0.3 \times 5.2 + 0.5) = 0.86$$

$$y_7' = \text{ReLU}(-2.2 \times 0.8 + 0.3 \times 3.7 + 1) = 0.35$$

$$S_5 = \text{Err}_5 \cdot y_5' = (y_{\text{actual}} - y_{\text{predicted}}) \cdot y_5' = (0 - 1.37) \times 1 = -1.37$$

here, if ReLU of y_5 is positive, then $y_5' = 1$, if negative, $y_5' = 0$

$$S_6 = \text{Err}_6 \cdot y_6' = (1 - 0.86) \times 1 = 0.14$$

$$S_7 = \text{Err}_7 \cdot y_7' = (0 - 0.35) \times 1 = -0.35$$

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$$\begin{aligned}\delta_3 &= (W_{35}\delta_5 + W_{36}\delta_6 + W_{37}\delta_7) y_3' \\ &= (-0.1(-1.37) + 2.4 \times 0.14 + (-2.2)(-0.35)) \times 1 \\ &= 1.243\end{aligned}$$

$$\begin{aligned}\delta_4 &= (W_{45}\delta_5 + W_{46}\delta_6 + W_{47}\delta_7) y_4' \\ &= (1.5 \times (-1.37) + (-5.2) \times 0.14 + 3.7 \times (-0.35)) \times 1 \\ &= 4.078\end{aligned}$$

$$W_{35} = W_{35} + \eta y_3 \delta_5 = -0.1 + 0.1 \times 0.8 \times (-1.37) = -0.21$$

$$W_{36} = W_{36} + \eta y_3 \delta_6 = 2.4 + 0.1 \times 0.8 \times 0.14 = 2.41$$

$$W_{45} = W_{45} + \eta y_4 \delta_5 = 1.5 + 0.1 \times 0.3 \times (-1.37) = 1.4589$$

$$W_{37} = W_{37} + \eta y_3 \delta_7 = -2.2 + 0.1 \times 0.8 \times (-0.35) = -2.23$$

$$W_{46} = W_{46} + \eta y_4 \delta_6 = -5.2 + 0.1 \times 0.3 \times 0.14 = -5.19$$

$$W_{47} = W_{47} + \eta y_4 \delta_7 = 3.7 + 0.1 \times 0.3 \times (-0.35) = 3.69$$

$$b_5 = b_5 + \eta \delta_5 = 1 + 0.1 \times (-1.37) = 0.863$$

$$b_6 = b_6 + \eta \delta_6 = 0.5 + 0.1 \times 0.14 = 0.514$$

$$b_7 = b_7 + \eta \delta_7 = 1 + 0.1 \times (-0.35) = 0.965$$

$$W_{13} = W_{13} + \eta x_{13} \delta_3 = -2.5 + 0.1 \times 0.5 \times 1.243 = -2.4$$

$$W_{14} = W_{14} + \eta x_{14} \delta_4 = -1.5 + 0.1 \times 0.5 \times 4.078 = -1.3$$

$$W_{23} = W_{23} + \eta x_{23} \delta_3 = 0.6 + 0.1 \times 0.75 \times 1.243 = 0.69$$

$$W_{24} = W_{24} + \eta x_{24} \delta_4 = 0.4 + 0.1 \times 0.75 \times 4.078 = 0.423$$

$$b_3 = b_3 + \eta \delta_3 = 1.6 + 0.1 \times 1.243 = 1.72$$

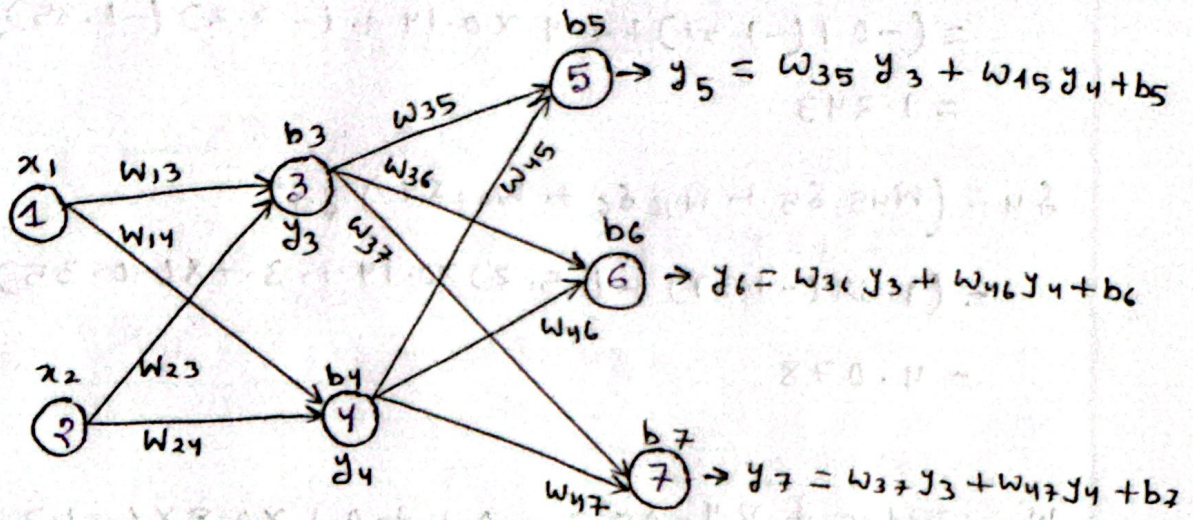
$$b_4 = b_4 + \eta \delta_4 = 0.75 + 0.1 \times 4.078 = 1.15$$

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Parameter Update without activation function in output layer



$$Loss = \frac{1}{2} \{ (ob_5 - y_5)^2 + (ob_6 - y_6)^2 + (ob_7 - y_7)^2 \}$$

$$\frac{dLoss}{dy_5} = \frac{1}{2} \cdot 2 (ob_5 - y_5) (-1) = -Err_5$$

$$\frac{dLoss}{dy_6} = \frac{1}{2} \cdot 2 (ob_6 - y_6) (-1) = -Err_6$$

$$\frac{dLoss}{dy_7} = \frac{1}{2} \cdot 2 (ob_7 - y_7) (-1) = -Err_7$$

$$y_5 = w_{35} y_3 + w_{45} y_4 + b_5$$

$$y_6 = w_{36} y_3 + w_{46} y_4 + b_6$$

$$y_7 = w_{37} y_3 + w_{47} y_4 + b_7$$

In case of sigmoid function, if $y_5 = 0.3 = \frac{1}{1 + e^{-z}}$

$$y_5' = y_5 (1 - y_5) = 0.3 (1 - 0.3)$$

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$$\frac{dLoss}{dW_{35}} = \frac{dLoss}{dY_5} \cdot \frac{dY_5}{dW_{35}} = -Err_5 \cdot \frac{d}{dW_{35}} (W_{35}Y_3 + W_{45}Y_4 + b_5)$$

$$= -Err_5 \cdot Y_3$$

$$\frac{dLoss}{dW_{45}} = \frac{dLoss}{dY_5} \cdot \frac{dY_5}{dW_{45}} = -Err_5 Y_4$$

$$\frac{dLoss}{dW_{36}} = \frac{dLoss}{dY_6} \cdot \frac{dY_6}{dW_{36}} = -Err_6 Y_3$$

$$\frac{dLoss}{dW_{46}} = \frac{dLoss}{dY_6} \cdot \frac{dY_6}{dW_{46}} = -Err_6 Y_4$$

$$\frac{dLoss}{dW_{37}} = \frac{dLoss}{dY_7} \cdot \frac{dY_7}{dW_{37}} = -Err_7 Y_3$$

$$\frac{dLoss}{dW_{47}} = \frac{dLoss}{dY_7} \cdot \frac{dY_7}{dW_{47}} = -Err_7 Y_4$$

$$\frac{dLoss}{db_5} = \frac{dLoss}{dY_5} \cdot \frac{dY_5}{db_5} = -Err_5$$

$$\frac{dLoss}{db_6} = \frac{dLoss}{dY_6} \cdot \frac{dY_6}{db_6} = -Err_6$$

$$\frac{dLoss}{db_7} = \frac{dLoss}{dY_7} \cdot \frac{dY_7}{db_7} = -Err_7$$

$$W_{35} = W_{35} + \eta Y_3 Err_5$$

$$W_{45} = W_{45} + \eta Y_4 Err_5$$

$$W_{36} = W_{36} + \eta Y_3 Err_6$$

$$W_{46} = W_{46} + \eta Y_4 Err_6$$

$$W_{37} = W_{37} + \eta Y_3 Err_7$$

$$W_{47} = W_{47} + \eta Y_4 Err_7$$

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$$b_5 = b_5 + \eta \text{Err}_5$$

$$b_6 = b_6 + \eta \text{Err}_6$$

$$b_7 = b_7 + \eta \text{Err}_7$$

$$Y_3 = \text{ReLU} \{ W_{13} X_1 + W_{23} X_2 + b_3 \}$$

$$Y_4 = \text{ReLU} \{ W_{14} X_1 + W_{24} X_2 + b_4 \}$$

$$\frac{d\text{Loss}}{dY_3} = \frac{d\text{Loss}}{dY_5} \frac{dY_5}{dY_3} + \frac{d\text{Loss}}{dY_6} \frac{dY_6}{dY_3} + \frac{d\text{Loss}}{dY_7} \frac{dY_7}{dY_3}$$

$$= (-\text{Err}_5) W_{35} + (-\text{Err}_6) W_{36} + (-\text{Err}_7) W_{37}$$

$$\frac{d\text{Loss}}{dY_4} = \frac{d\text{Loss}}{dY_5} \frac{dY_5}{dY_4} + \frac{d\text{Loss}}{dY_6} \frac{dY_6}{dY_4} + \frac{d\text{Loss}}{dY_7} \frac{dY_7}{dY_4}$$

$$= (-\text{Err}_5) W_{45} + (-\text{Err}_6) W_{46} + (-\text{Err}_7) W_{47}$$

$$\frac{d\text{Loss}}{dW_{13}} = \frac{d\text{Loss}}{dY_3} \frac{dY_3}{dW_{13}} = \{ (-\text{Err}_5) W_{35} + (-\text{Err}_6) W_{36} + (-\text{Err}_7) W_{37} \} \text{ReLU}'(W_{13} X_1 + W_{23} X_2 + b_3) X_1$$

$$\frac{d\text{Loss}}{dW_{23}} = \frac{d\text{Loss}}{dY_3} \frac{dY_3}{dW_{23}} = \{ (-\text{Err}_5) W_{35} + (-\text{Err}_6) W_{36} + (-\text{Err}_7) W_{37} \} \text{ReLU}'(W_{13} X_1 + W_{23} X_2 + b_3) X_2$$

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$$\frac{dLoss}{dW_{14}} = \frac{dLoss}{dY_4} \frac{dY_4}{dW_{14}} = \{ (-Err_5) W_{45} + (-Err_6) W_{46} + (-Err_7) W_{47} \} \text{ReLU}'(W_{14}X_1 + W_{24}X_2 + b_4) X_1$$

$$\frac{dLoss}{dW_{24}} = \frac{dLoss}{dY_4} \frac{dY_4}{dW_{24}} = \{ (-Err_5) W_{45} + (-Err_6) W_{46} + (-Err_7) W_{47} \} \text{ReLU}'(W_{14}X_1 + W_{24}X_2 + b_4) X_2$$

$$\frac{dLoss}{db_3} = \frac{dLoss}{dY_3} \frac{dY_3}{db_3} = \{ (-Err_5) W_{35} + (-Err_6) W_{36} + (-Err_7) W_{37} \} \text{ReLU}'(W_{13}X_1 + W_{23}X_2 + b_3)$$

$$\frac{dLoss}{db_4} = \frac{dLoss}{dY_4} \frac{dY_4}{db_4} = \{ (-Err_5) W_{45} + (-Err_6) W_{46} + (-Err_7) W_{47} \} \text{ReLU}'(W_{14}X_1 + W_{24}X_2 + b_4)$$

$$W_{13} = W_{13} + \eta \{ Err_5 W_{35} + Err_6 W_{36} + Err_7 W_{37} \} \text{ReLU}'(W_{13}X_1 + W_{23}X_2 + b_3) X_1$$

$$W_{23} = W_{23} + \eta \{ Err_5 W_{35} + Err_6 W_{36} + Err_7 W_{37} \} \text{ReLU}'(W_{13}X_1 + W_{23}X_2 + b_3) X_2$$

$$W_{14} = W_{14} + \eta \{ Err_5 W_{45} + Err_6 W_{46} + Err_7 W_{47} \} \text{ReLU}'(W_{14}X_1 + W_{24}X_2 + b_4) X_1$$

$$W_{24} = W_{24} + \eta \{ Err_5 W_{45} + Err_6 W_{46} + Err_7 W_{47} \} \text{ReLU}'(W_{14}X_1 + W_{24}X_2 + b_4) X_2$$

$$b_3 = b_3 + \eta \{ Err_5 W_{35} + Err_6 W_{36} + Err_7 W_{37} \} \text{ReLU}'(W_{13}X_1 + W_{23}X_2 + b_3)$$

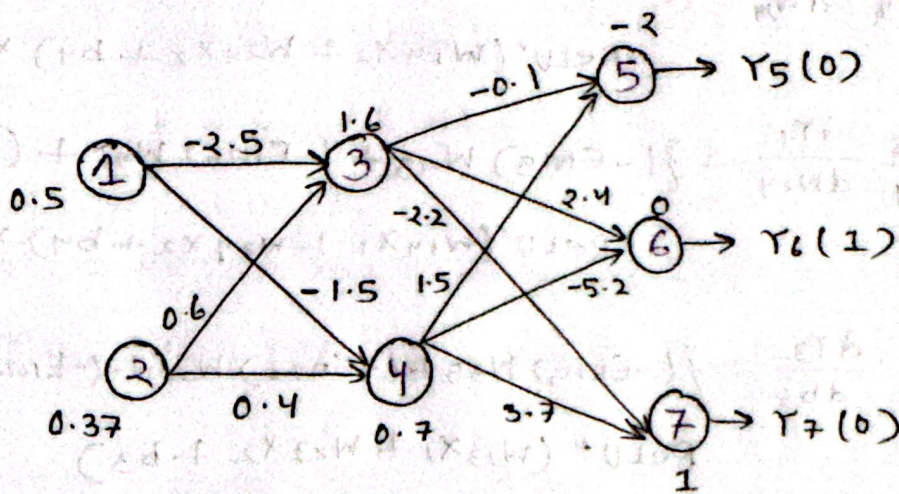
$$b_4 = b_4 + \eta \{ Err_5 W_{45} + Err_6 W_{46} + Err_7 W_{47} \} \text{ReLU}'(W_{14}X_1 + W_{24}X_2 + b_4)$$

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* Normal function in output layer and ReLU in hidden layer



$$Y_3 = \text{ReLU}(W_{13}X_1 + W_{23}X_2 + b_3)$$

$$= \text{ReLU}(0.5(-2.5) + 0.6 \times 0.37 + 1.6) = 0.57$$

$$Y_4 = \text{ReLU}(W_{14}X_1 + W_{24}X_2 + b_4)$$

$$= \text{ReLU}(0.5(-1.5) + 0.37 \times 0.4 + 0.7) = 0.1$$

$$Y_5 = W_{35}Y_3 + W_{45}Y_4 + b_5 = 0.57(-0.1) + 0.1 \times 1.5 - 2 = -1.9$$

$$Y_6 = W_{36}Y_3 + W_{46}Y_4 + b_6 = 2.4 \times 0.57 + 0.1(-5.2) + 0 = 0.85$$

$$Y_7 = W_{37}Y_3 + W_{47}Y_4 + b_7 = 0.57(-2.2) + 0.1 \times 3.7 + 1 = 0.12$$

$$\text{Err}_5 = Y_{\text{actual}} - Y_{\text{predicted}} = 0 - (-1.9) = 1.9$$

$$\text{Err}_6 = 1 - 0.85 = 0.15$$

$$\text{Err}_7 = 0 - 0.12 = -0.12$$

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$$W_{35} = W_{35} + \eta Y_3 E_{m5}$$

$$= -0.1 + 0.1 \times 0.57 \times 1.9 = 0.01$$

$$W_{46} = W_{46} + \eta Y_4 E_{m6}$$

$$= -5.2 + 0.1 \times 0.1 \times 0.15 = -5.19$$

$$b_5 = b_5 + \eta E_{m5} = -2 + 0.1 \times 1.9 = -1.81$$

$$b_6 = b_6 + \eta E_{m6} = 0 + 0.1 \times 0.15 = 0.015$$

$$b_7 = b_7 + \eta E_{m7} = 1 + 0.1 \times (-0.12) = 0.99$$

$$W_{13} = W_{13} + \eta (E_{m5} W_{35} + E_{m6} W_{36} + E_{m7} W_{37})$$

$$\text{ReLU}'(W_{13}X_1 + W_{23}X_2 + b_3)X_1$$

$$= -2.5 + 0.1 \times ((-0.12) \times 1.9 + 2.4 \times 0.15 + (-2.2) \times (-0.12))$$

$$\times \text{ReLU}'(0.57) \times 0.5 = -2.28$$

$$W_{23} = W_{23} + \eta (E_{m5} W_{35} + E_{m6} W_{36} + E_{m7} W_{37})$$

$$\text{ReLU}'(W_{13}X_1 + W_{23}X_2 + b_3)X_2$$

$$= 0.6 + 0.1 (1.9(-0.1) + 0.15 \times 2.4 + (-0.12) \times (-2.2))$$

$$\text{ReLU}'(0.57) \times 0.37$$

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$$W_{14} = W_{14} + \eta (E_{nn5} W_{n5} + E_{nn6} W_{n6} + E_{nn7} W_{n7})$$

$$\text{ReLU}'(W_{14}x_1 + W_{24}x_2 + b_4) x_1$$

$$= -1.5 + 0.1(1.9 \times 1.5 + 0.15(-5.2) + (-0.12)(3.7))$$

$$\text{ReLU}'(0.1) 0.5$$

$$W_{24} = W_{24} + \eta (E_{nn5} W_{n5} + E_{nn6} W_{n6} + E_{nn7} W_{n7})$$

$$\text{ReLU}'(W_{14}x_1 + W_{24}x_2 + b_4) x_2$$

$$= 0.4 + 0.1(1.5 \times 1.9 + (-5.2)0.15 + 3.7(-0.12))$$

$$\text{ReLU}'(0.1) 0.37 = 1.0$$

$$b_3 = b_3 + \eta (E_{nn5} W_{n5} + E_{nn6} W_{n6} + E_{nn7} W_{n7})$$

$$\text{ReLU}'(W_{13}x_1 + W_{23}x_2 + b_3)$$

$$= 1.6 + 0.1(1.9 \times (-0.1) + 0.15 \times 2.4 + (-0.12)(-2.2))$$

$$\text{ReLU}'(0.57)$$

$$b_4 = b_4 + \eta (E_{nn5} W_{n5} + E_{nn6} W_{n6} + E_{nn7} W_{n7})$$

$$\text{ReLU}'(W_{14}x_1 + W_{24}x_2 + b_4)$$

$$= 0.7 + 0.1(1.9 \times 1.5 + 0.15(-5.2) + (-0.12)(3.7))$$

$$\text{ReLU}'(0.1)$$

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$$W_{36} = W_{36} + \eta \cdot Em_6 Y_3$$

$$= 2.4 + 0.1 \times 0.15 \times 0.57 = 2.41$$

$$W_{37} = W_{37} + \eta Em_7 Y_3$$

$$= -2.2 + 0.1 \times (-0.12) \times 0.57 = -2.21$$

$$W_{45} = W_{45} + \eta Em_5 Y_4$$

$$= 1.5 + 0.1 \times 1.9 \times 0.1 = 1.52$$

$$W_{47} = W_{47} + \eta Em_7 Y_4$$

$$= 3.7 + 0.1 \times (-0.12) \times 0.1 = 3.699$$